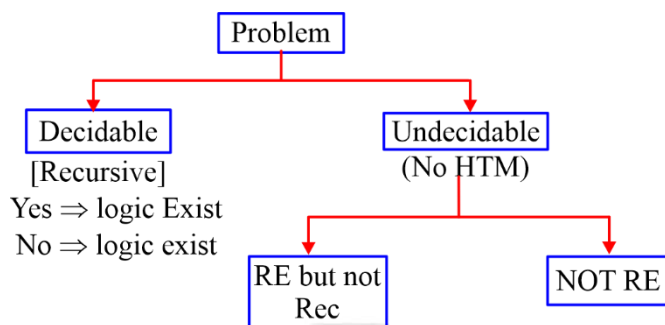


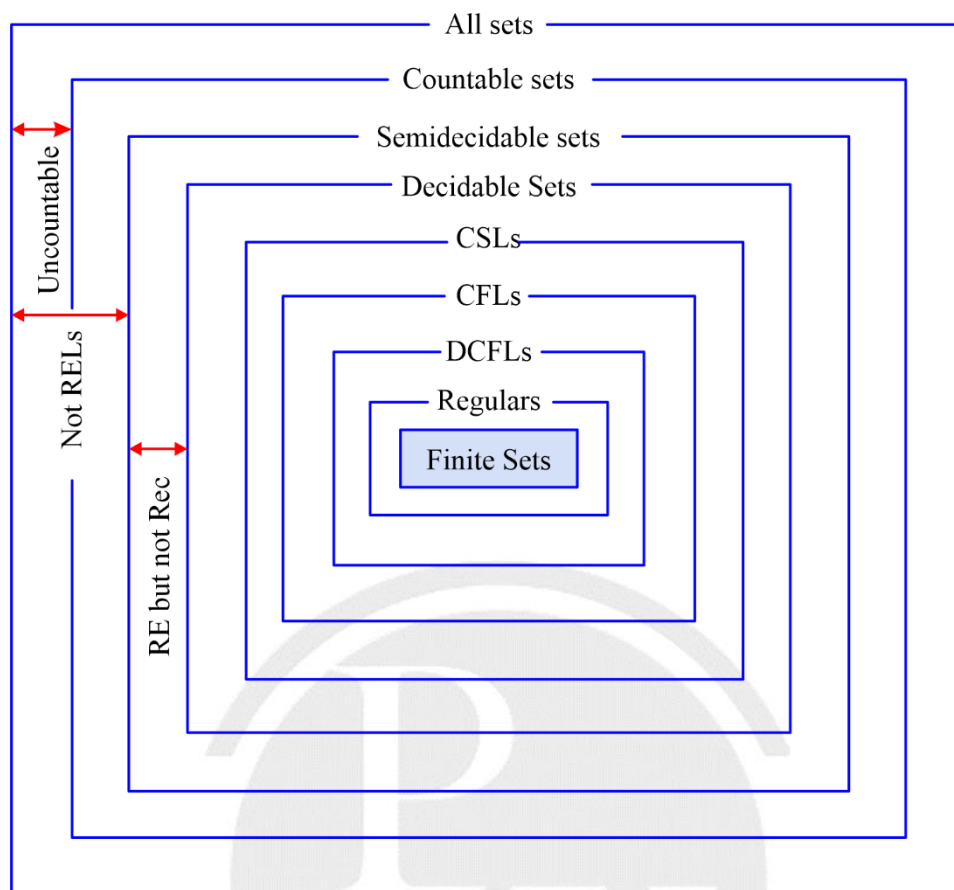
- Complement of Recursive set is Recursive.
- Complement of REL is either Recursive or non-REL.
- Complement of REL never be “REL which is not recursive”.

4.13 Decidable and Undecidable



4.14 Decidable Vs Undecidable, RE Vs Not RE, Countable Vs Uncountable

[1]	Decidable	and	Undecidable
	↓		↓
	HTM exist (Rec language)		HTM not exist (not Rec language)
[2]	RE	and	Not RE
	↓		↓
	TM exist		TM not exist
[3]	Countable set	and	Uncountable set
	↓		↓
	X is countable set iff X → f(X) is bijective where f(X) is known countable set.		(X is not countable set)



Note :

- If problem p is decidable then $\bar{p}$ is also decidable.
- If problem p is Undecidable then $\bar{p}$ is also UD.
- If problem p is RE but not recursive then $\bar{p}$ is not RE.
- If problem p is not RE then $\bar{p}$ is either “Not RE” or “RE but not Recursive”.

4.15 Decision Properties Table

- D: Decidable
- UD: Undecidable

	FA	DPDA	PDA	LBA/HTM	TM
H (Halting)	D	D	D	D	UD
M (Membership)	D	D	D	D	UD
E_m (Emptiness)	D	D	D	UD	UD
F (Finiteness)	D	D	D	UD	UD

T (Totality)	D	D	UD	UD	UD
E_q (Equivalence)	D	D	UD	UD	UD
D (Disjoint)	D	UD	UD	UD	UD
S (Set Containment)	D	UD	UD	UD	UD

4.16 Decidable problem for DFA / NFA/ FA/ Regular

- (1) Halting problem for FA / Reg/ DFA / NFA is decidable.
- (2) Non-halting problem for FA / Reg/ DFA / NFA is decidable.
- (3) Membership problem for FA / Reg/ DFA / NFA is decidable.
- (4) Non-membership problem for FA / Reg/ DFA / NFA is decidable.
- (5) Emptiness for FA / Reg/ DFA / NFA is decidable.
- (6) Non-emptiness problem for FA / Reg/ DFA / NFA is decidable.
- (7) Fitness problem for FA / Reg/ DFA / NFA is decidable.
- (8) Non-fitness problem for FA / Reg/ DFA / NFA is decidable.
- (9) Totality problem for FA / Reg/ DFA / NFA is decidable.
- (10) Non-totality problem for FA / Reg/ DFA / NFA is decidable.
- (11) Equivalence problem for FA / Reg/ DFA / NFA is decidable.
- (12) Non-equivalence problem for FA / Reg/ DFA / NFA is decidable.
- (13) Disjointness problem is decidable for FA / Reg/ DFA / NFA.
- (14) Non-disjointness problem is decidable for FA / Reg/ DFA / NFA.
- (15) Set containment problem for FA / Reg/ DFA / NFA is decidable.
- (16) Non-set containment problem for FA / Reg/ DFA / NFA is decidable.

4.17 Decidable problems for CFLs/DCFLs

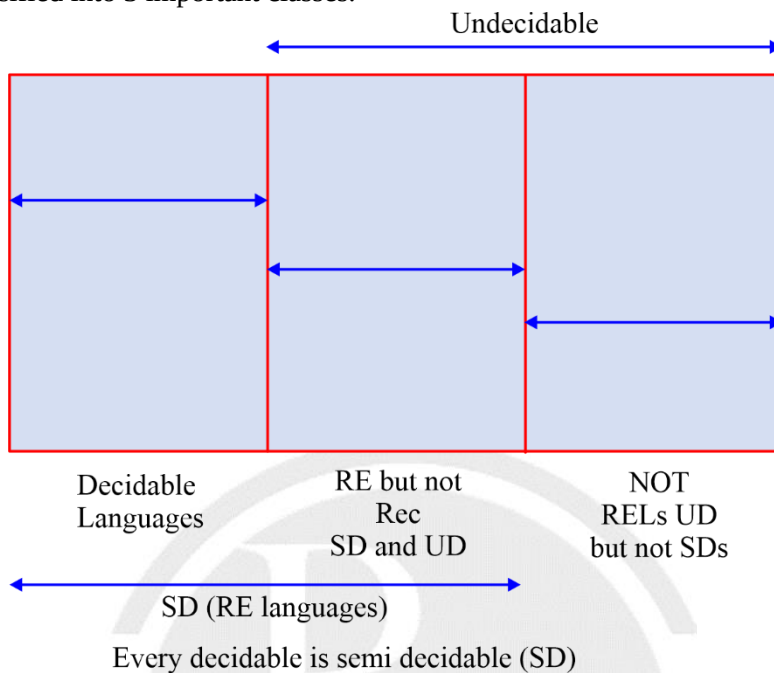
	Problems	DCFLs	CFLs
	Halting	D	D
	Non-Halting	D	D
CYK algo {	Membership	D	D
	Non-membership	D	D
Simplification algo {	Emptiness	D	D
	Non-emptiness	D	D
Dependancy graph {	Finiteness	D	D
	Non-finiteness	D	D
	Totality	D	UD
	Non-totality	D	UD
	Equivalence	D	UD
	Disjointness	UD	UD
	Non-disjointness	UD	UD
	Set containment	UD	UD
	Non-set containment	UD	UD

4.18 Decidability problems for Recursive languages

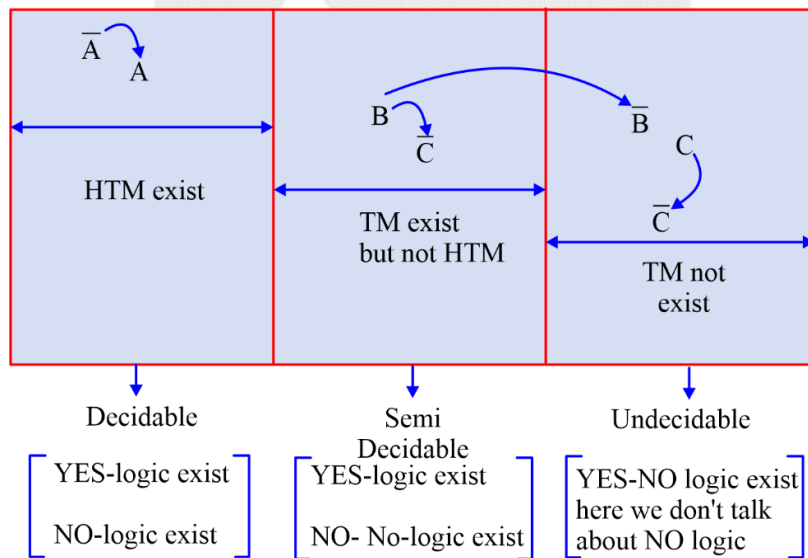
	Problems	Recursive
1.	Halting	D
2.	Non-Halting	D
3.	Membership	D
4.	Non-membership	D
5.	Emptiness	UD [Not REL]
6.	Non-emptiness	UD [RE but not Rec] [SD but UD]
7.	Finiteness	UD [Not RE]
8.	Non-finiteness	UD [Not RE]
9.	Totality	UD [Not RE]
10.	Non-totality	UD [RE but not Rec] [SD but UD]
11.	Equivalence	UD [Not RE]
12.	Non-equivalence	UD [RE but not Rec]
13.	Disjointness	UD [Not RE]
14.	Non-disjointness	UD [RE but not Rec]
15.	Set containment	UD [Not RE]
16.	Non-set containment	UD [RE but not Rec]

4.19 Classification of Languages based on Decidability

All RE languages can be classified into 3 important classes.



4.19.1 Decidability Vs Turing Machine



4.20 Decidable languages

- (1) Finite set $\Rightarrow$ Decidable
- (2) $\Sigma = \{a, b\} \Rightarrow$ Decidable
- (3) $\Sigma \cong$ Set of finite number of symbols $\Rightarrow$ Decidable
- (4) Σ^* over alphabet $\Sigma = \{a, b\} \Rightarrow$ Decidable
- (5) $\{M \mid M \text{ is DFA, } M \text{ accepts } ab\} \Rightarrow$ Decidable



- (6) $\{TM \mid \text{Number of states in TM} = 2\} \Rightarrow \text{Decidable}$
- (7) $\{TM \mid \text{TM reaches state } q \text{ within 100 steps}\} \Rightarrow \text{Decidable}$
- (8) $\{TM \mid \text{TM accepts REL}\} \Rightarrow \text{Decidable}$

□□□□

